

Chaotic Amplitude Control for the Ising Minimization using Optical Parametric Oscillator Systems - Executive Summary

Combinatorial optimization problems, often NP-hard, are widespread in fields such as drug discovery and finance. Traditional digital computing struggles with these complexities, prompting a shift towards sustainable computational paradigms. This includes analog, physics-based algorithms and hardware like the Ising and XY models, leveraging principles such as second-order phase transitions and bistability in systems like lasers and degenerate optical parametric oscillators (DOPOs). The Coherent Ising Machine (CIM) exemplifies this approach, mapping optimization problems onto Ising model ground-state searches. This thesis explores CIM principles, parallel processing potential, and addresses performance challenges. It also evaluates novel CIM methodologies on the MAX-CUT problem using varied graph types, aiming to advance quantum-optical computing for NP-hard problems and validate claims of analog system efficiency in solving the Ising Hamiltonian.

Significance and Interest of the project

This project is intriguing for its interdisciplinary approach: tackling the limitations of traditional computing in solving complex optimization problems by combining quantum optics, nonlinear dynamics, and computational mathematics. CIMs offer a novel hardware paradigm drawing from quantum mechanics and optical physics, potentially surpassing classical algorithms. This could revolutionize computational-intensive fields, advancing sustainable and powerful computing technologies.

What was done in the project?

This project extensively investigated Coherent Ising Machines (CIMs) with a primary focus on optimizing their performance through chaotic amplitude control for solving the Ising minimization problem. The research began by establishing the foundational mathematical framework of CIMs, detailing their methodology for mapping complex optimization problems onto the Ising model while identifying critical performance challenges such as freeze-out and amplitude heterogeneity. Various advanced feedback control strategies including nonlinear, chaotic amplitude, and separable feedback controls were explored to enhance CIM performance by dynamically adjusting system parameters. Numerical simulations rigorously analyzed different CIM configurations under these strategies, evaluating similarity metrics, correlation properties, and trajectory variances to assess their impact on solving optimization problems. Additionally, the project investigated methods to reduce energy consumption in CIM systems, crucial for sustainable computing solutions. Practical experiments using numerical integration techniques confirmed that chaotic amplitude control consistently yielded superior results compared to other methods, despite challenges like susceptibility to local minima, suggesting avenues for further optimization and parameter exploration in future research.

The thesis concludes by summarizing the findings and emphasizing the potential of CIMs in solving combinatorial optimization problems more efficiently than traditional methods. The exploration of chaotic amplitude control is highlighted as a promising direction for future research, offering new avenues for improving the performance and scalability of CIMs. The work contributes to the broader goal of developing innovative computational paradigms that leverage physical principles to overcome the limitations of classical computing.